Two functions f and g are said to be ***equal functions*** if their domains are same and $f(a) = g(a)$ for all a in the domain.

If f is a function with domain A and co-domain B, we write $f : A \to B$ (Read this as f is from A to B or f be a function from A to B). We also say that f maps A into B. If $f(a) = b$, then we say f maps a to b or a is mapped onto b by f, and so on.

The range of a function is the collection of all elements in the co-domain which have pre-images. Clearly the range of a function is a subset of the co-domain. Further the first condition says that every element in the domain must have an image; this is the reason for defining the domain of a relation R from a set A to a set B as the set of all elements of A having images and not as A. The second condition says that an element in the domain cannot have two or more images.

Naturally one may have the following doubts:

- In the definition, why we use the definite article "the" for image of a and the indefinite article "a" for pre-image of b?
- We have a condition stating that every element in the domain must have an image; is there any condition like "every element in the co-domain must have a pre-image"? If not, why?
- We have a condition stating that an element in the domain cannot have two or more images; is there any condition like "an element in the co-domain cannot have two or more pre-images"? If not, why?

As an element in the domain has exactly one image and an element in the co-domain can have more than one pre-image according to the definition, we use the definite article "the" for image of a and the indefinite article "a" for pre-image of b. There are no conditions as asked in the other two questions; the reason behind it can be understood from the problem of students' mark we considered above.

We observe that every function is a relation but a relation need not be a function.

Let $f = \{(a, 1), (b, 2), (c, 2), (d, 4)\}$.

Is f a function? This is a function from the set $\{a, b, c, d\}$ to $\{1, 2, 4\}$. This is not a function from $\{a, b, c, d, e\}$ to $\{1, 2, 3, 4\}$ because e has no image. This is not a function from $\{a, b, c, d\}$ to $\{1, 2, 3, 5\}$ because the image of d is not in the co-domain; f is not a subset of $\{a, b, c, d\} \times \{1, 2, 3, 5\}$. So whenever we consider a function the domain and the co-domain must be stated explicitly.

The relation discussed in Illustration 1.1 is a function with domain $\{L, E, T, U, S, W, I, N\}$ and co-domain $\{O, H, W, X, V, Z, L, Q\}$. The relation discussed in Illustration 1.3 is again a function with domain $\{a, b, \ldots, z\}$ and the co-domain $\{1, 2, 3, \ldots, 26\}$.

In Illustration 1.2, we discussed three relations, namely

$$\text{(i) } y = 2x \quad \text{(ii) } y = x^2 \quad \text{(iii) } y^2 = x.$$

Clearly (i) and (ii) are functions whereas (iii) is not a function, if the domain and the co-domain are $\mathbb{R}$. In (iii) for the same x, we have two y values which contradict the definition of the function. But if we split into two relations, that is, $y = \sqrt{x}$ and $y = -\sqrt{x}$ then both become functions with same domain non-negative real numbers and the co-domains $[0, \infty)$ and $(-\infty, 0]$ respectively.

1.6.1 Ways of Representing Functions

(a) Tabular Representation of a Function

When the elements of the domain are listed like $x_1, x_2, x_3 \ldots x_n$, we can use this tabular form.Here, the values of the arguments $x_1, x_2, x_3 \ldots x_n$ and the corresponding values of the function $y_1, y_2, y_3 \ldots y_n$ are written out in a definite order.

x	x_1	x_2	$\ldots$	x_n
y	y_1	y_2	$\ldots$	y_n

(b) Graphical Representation of a Function

When the domain and the co-domain are subsets of $\mathbb{R}$, many functions can be represented using a graph with x-axis representing the domain and y-axis representing the co-domain in the (x, y)-plane.